Session 5 – Assignment Answers

1. Write the Git command for each task.

Task	Git Command
Initialize a repository	`git init`
Stage all files	`git add .`
Commit with a message	`git commit -m "Initial commit"`
Push to remote	`git push origin main`
Pull latest changes	`git pull origin main`

2. What is the difference between `git fetch` and `git pull`? Answer in 2 lines.

- `git fetch`: Downloads the latest changes from the remote repository but does not merge them into the current branch.
- `git pull`: Downloads the latest changes and integrates them into the current branch, usually by merging or rebasing.

3. Create a new GitHub repository, push any small text file from your local system, and share the repo link.

Repository URL:
`https://github.com/sumitttsoni/C_Language/blob/main/Assignments/Session_1_Assignment_1.pdf`.

4. What is the difference between a Fork and a Pull Request? Answer in 2 lines.

- Fork: A fork creates a copy of another user's repository under your own GitHub account.
- Pull Request (PR): A pull request proposes changes from one branch or repository to another so they can be reviewed and merged.

5. Create a new branch named `feature-test`, make one commit on it, and merge it back to the main branch. Write the commands you used.

Run these commands inside your local Git repository:

Step 1: Create and switch to the new branch

```
git switch -c feature-test
```

Step 2: Create a file and make a commit

```
echo This is my feature test > feature-test.txt
```

```
git add .
git commit -m "Add feature test file"
```

Step 3: Switch back to the main branch

```
git switch main
```

Step 4: Merge the feature branch into main

```
git merge feature-test
```

Step 5: Push the changes to GitHub (optional)

```
git push origin main
git push -u origin feature-test
```

Expected result: The `feature-test.txt` file and its commit will be included in the `main` branch after a successful merge.